

Some Properties of the Sinkhorn Distance

Setting

Let $\alpha \geq 0$, and let $M \in \mathbb{R}_+^{d \times d}$ satisfy the following properties:

$$M_{ij} = 0 \iff i = j,$$

$$M_{ij} = M_{ji} \quad \text{for all } i, j,$$

and

$$M_{ik} \leq M_{ij} + M_{jk} \quad \text{for all } i, j, k.$$

Let $r, c \in \Sigma_d$. We define

$$\mathbf{U}(r, c) = \left\{ P \in \mathbb{R}_+^{d \times d} : P\mathbf{1} = r, \quad P^\top \mathbf{1} = c \right\}.$$

For $\alpha \geq 0$, we also define

$$\mathbf{U}_\alpha(r, c) = \{ P \in \mathbf{U}(r, c) : h(P) \geq h(r) + h(c) - \alpha \}.$$

The Sinkhorn distance is defined by

$$d_{M, \alpha}(r, c) = \min_{P \in \mathbf{U}_\alpha(r, c)} \langle P, M \rangle,$$

where

$$\langle P, M \rangle = \sum_{i, j} P_{ij} M_{ij}.$$

Symmetry

We first show that

$$d_{M, \alpha}(r, c) = d_{M, \alpha}(c, r).$$

Let $P \in \mathbf{U}_\alpha(r, c)$. By definition,

$$P\mathbf{1} = r, \quad P^\top \mathbf{1} = c, \quad h(P) \geq h(r) + h(c) - \alpha.$$

Then

$$P^\top \mathbf{1} = c, \quad (P^\top)^\top \mathbf{1} = P\mathbf{1} = r.$$

Moreover,

$$h(P^\top) = h(P).$$

Therefore,

$$P \in \mathbf{U}_\alpha(r, c) \iff P^\top \in \mathbf{U}_\alpha(c, r).$$

Now, since $M = M^\top$, we have

$$\begin{aligned}
\langle P^\top, M \rangle &= \langle P^\top, M^\top \rangle \\
&= \sum_{i,j} (P^\top)_{ij} (M^\top)_{ij} \\
&= \sum_{i,j} P_{ji} M_{ji} \\
&= \sum_{i,j} P_{ij} M_{ij} \\
&= \langle P, M \rangle.
\end{aligned}$$

Thus the two minimization problems have the same set of values, and therefore

$$d_{M,\alpha}(r, c) = d_{M,\alpha}(c, r).$$

Triangle Inequality

Let $x, y, z \in \Sigma_d$.

Let

$$P \in \mathcal{U}_\alpha(x, y), \quad Q \in \mathcal{U}_\alpha(y, z).$$

By the gluing lemma, there exists

$$S \in \mathcal{U}_\alpha(x, z)$$

such that, for every i, k ,

$$S_{ik} = \sum_{\substack{j \\ y_j > 0}} \frac{P_{ij} Q_{jk}}{y_j}.$$

Let

$$P^\star \in \arg \min_{P \in \mathcal{U}_\alpha(x, y)} \langle P, M \rangle$$

and

$$Q^\star \in \arg \min_{Q \in \mathcal{U}_\alpha(y, z)} \langle Q, M \rangle.$$

Define

$$S_{ik} = \sum_{\substack{j \\ y_j > 0}} \frac{P_{ij}^\star Q_{jk}^\star}{y_j}.$$

Then, by the previous lemma,

$$S \in \mathcal{U}_\alpha(x, z).$$

Hence, by definition of the Sinkhorn distance,

$$d_{M,\alpha}(x, z) \leq \langle S, M \rangle.$$

Now we compute:

$$\begin{aligned}
\langle S, M \rangle &= \sum_{i,k} S_{ik} M_{ik} \\
&= \sum_{i,k} \left(\sum_{\substack{j \\ y_j > 0}} \frac{P_{ij}^\star Q_{jk}^\star}{y_j} \right) M_{ik} \\
&= \sum_{i,j,k} \frac{P_{ij}^\star Q_{jk}^\star}{y_j} M_{ik}.
\end{aligned}$$

Since M satisfies the triangle inequality,

$$M_{ik} \leq M_{ij} + M_{jk}.$$

Therefore,

$$\begin{aligned} \langle S, M \rangle &\leq \sum_{i,j,k} \frac{P_{ij}^* Q_{jk}^*}{y_j} (M_{ij} + M_{jk}) \\ &= \sum_{i,j,k} \frac{P_{ij}^* Q_{jk}^*}{y_j} M_{ij} + \sum_{i,j,k} \frac{P_{ij}^* Q_{jk}^*}{y_j} M_{jk}. \end{aligned}$$

For the first term,

$$\sum_{i,j,k} \frac{P_{ij}^* Q_{jk}^*}{y_j} M_{ij} = \sum_{i,j} P_{ij}^* M_{ij} \left(\sum_k \frac{Q_{jk}^*}{y_j} \right).$$

Since $Q^* \in U_\alpha(y, z)$, we have

$$\sum_k Q_{jk}^* = y_j.$$

Hence

$$\sum_k \frac{Q_{jk}^*}{y_j} = 1,$$

and therefore

$$\sum_{i,j,k} \frac{P_{ij}^* Q_{jk}^*}{y_j} M_{ij} = \sum_{i,j} P_{ij}^* M_{ij} = \langle P^*, M \rangle = d_{M,\alpha}(x, y).$$

For the second term,

$$\sum_{i,j,k} \frac{P_{ij}^* Q_{jk}^*}{y_j} M_{jk} = \sum_{j,k} Q_{jk}^* M_{jk} \left(\sum_i \frac{P_{ij}^*}{y_j} \right).$$

Since $P^* \in U_\alpha(x, y)$, we have

$$\sum_i P_{ij}^* = y_j.$$

Thus

$$\sum_i \frac{P_{ij}^*}{y_j} = 1,$$

and therefore

$$\sum_{i,j,k} \frac{P_{ij}^* Q_{jk}^*}{y_j} M_{jk} = \sum_{j,k} Q_{jk}^* M_{jk} = \langle Q^*, M \rangle = d_{M,\alpha}(y, z).$$

We obtain

$$\langle S, M \rangle \leq d_{M,\alpha}(x, y) + d_{M,\alpha}(y, z).$$

Since

$$d_{M,\alpha}(x, z) \leq \langle S, M \rangle,$$

we conclude that

$$d_{M,\alpha}(x, z) \leq d_{M,\alpha}(x, y) + d_{M,\alpha}(y, z).$$

Thus $d_{M,\alpha}$ satisfies the triangle inequality.

A Modified Distance

We now define

$$D_{M,\alpha}(r, c) = \begin{cases} 0, & \text{if } r = c, \\ d_{M,\alpha}(r, c), & \text{if } r \neq c. \end{cases}$$

We show that

$$D_{M,\alpha}(r, c) = 0 \iff r = c.$$

If $r = c$, then by definition

$$D_{M,\alpha}(r, c) = 0.$$

Conversely, assume that

$$D_{M,\alpha}(r, c) = 0.$$

If $r \neq c$, then by definition

$$D_{M,\alpha}(r, c) = d_{M,\alpha}(r, c).$$

So we would have

$$d_{M,\alpha}(r, c) = 0.$$

Thus there exists $P \in U_\alpha(r, c)$ such that

$$\langle P, M \rangle = 0.$$

That is,

$$\sum_{i,j} P_{ij} M_{ij} = 0.$$

Since

$$P_{ij} \geq 0, \quad M_{ij} \geq 0,$$

and since

$$M_{ij} > 0 \quad \text{whenever } i \neq j,$$

we must have

$$P_{ij} = 0 \quad \text{for every } i \neq j.$$

Therefore P is diagonal.

Since $P \in U_\alpha(r, c) \subseteq U(r, c)$, we have

$$\sum_j P_{ij} = r_i, \quad \sum_i P_{ij} = c_j.$$

But P is diagonal, so

$$r_i = P_{ii} = c_i$$

for every i . Hence

$$r = c,$$

which contradicts the assumption $r \neq c$.

Therefore,

$$D_{M,\alpha}(r, c) = 0 \implies r = c.$$

We conclude that

$$D_{M,\alpha}(r, c) = 0 \iff r = c.$$

Conclusion

The modified Sinkhorn distance $D_{M,\alpha}$ satisfies:

$$D_{M,\alpha}(r, c) \geq 0,$$

$$D_{M,\alpha}(r, c) = 0 \iff r = c,$$

$$D_{M,\alpha}(r, c) = D_{M,\alpha}(c, r),$$

and

$$D_{M,\alpha}(x, z) \leq D_{M,\alpha}(x, y) + D_{M,\alpha}(y, z).$$

Thus $D_{M,\alpha}$ has the expected metric-type properties.